

INSTRUCTIONS

1. Submission is only via Themis. Deadlines are strict.
 2. The exercises in this assignment add up to 100 points. To calculate your grade simply divide the number of points by 10.
 3. Seeking solutions from the internet, from any external resource, or from any other person is prohibited.
 4. Please note that the course lecturer reserves the right to ask the student submitting the assignment to explain the answers to any or all questions. If the student is unable to provide a satisfactory answer then that question may receive partial/no credit.
 5. Of course, university policies on plagiarism always apply. In particular, any suspected plagiarism will be reported to the Board of Examiners.
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A note on how Themis works

For this assignment you will be using Themis in order to have your submissions automatically checked! Unlike previous courses you may have taken, there will be no style points which the TAs enforce, and you are free (and encouraged!) to come up with your own programming style.

That being said, for this lab in particular, since we will not be handling input via console yet, all input will be given to you through registers! In order for you to emulate this you can simply assemble, initialize your machine, and place the values manually inside of LC3 Tools!

For some exercises in this lab **efficiency is important** and will be taken into account! Therefore, if your program is not fast enough, you will fail with a **timeout** on Themis! Keep in mind that in on Themis we will measure timeouts by counting CPU cycles rather than time itself, so don't be deceived by how fast Themis moves!

1. Addition

Welcome to the first LC3 exercise! For this exercise you simply need to add two numbers together. Input for the first number will be passed to you through R0 and for the second number through R1. You are then expected to place the output within R2! **(10 points)**

Example 1**Input:**

R0: 5 R1: 8

Output:

R2: 13

Example 2**Input:**

R0: 10 R1: 8

Output:

R2: 18

2. Subtraction

This exercise is quite similar to the previous one. This time, instead of addition you are expected to perform subtraction! Input will be passed to you in the same manner, which is through R0 and R1 with output being expected in R2. Remember that LC3 works in 2s complement! (10 points)

Example 1

Input:

R0: 5
R1: 8

Output:

R2: -3

Example 2

Input:

R0: 10
R1: 8

Output:

R2: 2

3. Even Odd Checker

For this problem we are trying to see if a given input is even or odd! Input will be passed to you through R0. If the input is odd you should place 1 inside of R1, if the input is even, you should place 0 inside of R1! Keep in mind that there is no division / modulo operator in LC3! Similar to the previous exercise, remember that LC3 stores values in 2s complement. Does your method also work for negative numbers? (20 points)

Example 1

Input:

R0: 5

Output:

R1: 1

Example 2

Input:

R0: 10

Output:

R1: 0

4. Multiplication

This exercise is quite similar to the previous ones. This time, instead of addition or subtraction you have to implement multiplication! Input will be passed to you in the same manner, which is through R0 and R1 with output being expected in R2. Keep in mind that input could have any value, including negative ones, or zero for either one of the inputs. **(40 points)**

Example 1

Input:

```
R0: 5
R1: 8
```

Output:

```
R2: 40
```

Example 2

Input:

```
R0: -10
R1: 8
```

Output:

```
R2: -80
```

5. Calculator

We've spent quite a bit of time building addition, subtraction and multiplication! It is now time to combine all of our code into a single program, by building a calculator! The input for this problem will differ a bit from the previous problems. The first number will come from R1, and the second number will come from R2. Finally, R3 will be used to signal the operation being performed, with a value of -1 being used for subtraction, a value of 0 being used for addition and a value of 1 being used for multiplication. The output of the operation should then be placed into R4! **(20 points)**

Example 1 (addition)

Input:

```
R1: 20
R2: 10
R3: 0
```

Output:

```
R4: 30
```

Example 2 (multiplication)**Input:**

R1: 20 R2: 10 R3: 1

Output:

R4: 200

Example 3 (subtraction)**Input:**

R1: 20 R2: 10 R3: -1

Output:

R4: 10
